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Comparative Analysis of Various Feature Extraction Methods on IoT 2023

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Abstract

The Internet of Things (IoT) has emerged as a transformative technology with applications spanning diverse domains, from healthcare to smart cities. Efficient utilization of IoT data is essential for extracting meaningful insights and supporting decision-making processes. This research paper focuses on the critical task of feature extraction from IoT data, with a particular emphasis on the IoT 2023 dataset—a comprehensive and widely-used benchmark dataset in the IoT domain. In this study, exploration and comparison of various feature extraction methods are undertaken to enhance understanding of IoT 2023 dataset characteristics and its potential applications. Traditional statistical techniques, as well as machine learning-based approaches for feature extraction, are investigated. Evaluation of these methods is based on their ability to capture relevant information, reduce dimensionality, and enhance the performance of downstream IoT analytics tasks. Extensive experiments are conducted, and a comprehensive performance analysis is presented to guide the selection of the most suitable feature extraction approach for specific IoT applications. The outcomes of this research contribute to ongoing efforts to harness the potential of IoT data by offering a comprehensive understanding of the impact of feature extraction methods on data quality and predictive accuracy. This aids in paving the way for more informed and effective IoT solutions in 2023 and beyond.

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Keywords: The Internet of Things 2023 Benchmark; Feature Evaluation; Dimensionality reduction; Comparative Analysis component

1. Introduction

The term ”Internet of Things” (IoT) describes a network of physical objects, or ”things,” that are integrated with different software, sensors, and other technologies. This allows the ”things” to gather and share data with other Internet-connected devices and systems. These objects can be commonplace things like automobiles, wearable technology, industrial machinery, and home appliances, among others. IoT devices gather data that can be utilised for automation,

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control, automation, and process insight.In the late 1990s, the phrase ”Internet of Things” was first used. Although there is some disagreement regarding the term’s precise origin, Kevin Ashton, a British technology pioneer and co-founder of the MIT Auto-ID Centre, is frequently credited with coining it. In 1999, he coined the term in a presentation to Procter & Gamble to explain a system that would allow items to have RFID tags attached in order to track inventory and enhance supply chain management. Since then, inventory management has not been the only industry or application that the idea of the Internet of Things has expanded to include. IoT (Internet of Things) devices are significantly threatened by malware, which can harm them by taking advantage of their vulnerabilities. Sensitive data such as login credentials, personal information, and any data recorded or saved by cameras, sensors, or other sensors, can be stolen by malware from Internet of Things devices. Botnets frequently aim to include Internet of Things devices. Many IoT devices can become infected with malware, which can then take control of them and turn them into a network of “zombie” devices that can be used for other malicious purposes, such as distributed denial of service (DDoS) assaults. Entry points into local networks can be provided by IoT devices. IoT device malware has the ability to change course, compromise other networked devices or obtain private information. Today a variety of machine learning algorithms are available to help detect which Internet of Things devices are susceptible to malware assaults. In the Internet of Things (IoT) space, feature extraction is essential because it makes it possible to handle, analyse, and understand IoT data more effectively and efficiently. IoT datasets can be quite large and complex, especially when they include data from sensors, devices, and linked systems. The term “curse of dimensionality” refers to the explosive tendencies of growing data dimensions. When this phenomena occurs, processing and analysis of it usually need more computational work. The Hughes Phenomenon also known as the curse of dimensionality says that one can achieve the ideal number of features, the classifier’s performance grows as the number of features increases as well. The performance of the classifier will then deteriorate when more features based on the same size as the training set are added. Feature extraction decreases the dimensionality of these datasets by selecting or designing relevant features, which makes them simpler to handle and need less computing power to process. feature extraction can be compared with medications as not all patients can take a medication this can be related with features as medications and patients as the dataset to be explored . In this paper, various feature extraction techniques are performed and their output is compared.

2. Related Area of Work

One of the more recent innovations in the Internet space is the Internet of Things. It is characterized as a network of interconnected physical and digital sensors and devices that produce and share vast volumes of data without the assistance of a human. The Internet of Things (IoT) can process more data faster and more efficiently than ever before since it does not require human operators. By examining the applicability of machine learning methods in the identification of irregularities discovered in the data of such networks, this research focuses on the security element of Internet of Things networks. It looks at machine learning algorithms that work well in scenarios that are somewhat similar and makes comparisons using a variety of parameters and techniques. The following algorithms are implemented in this paper [1] by the author (Stoian, 2020): AdaBoost (ADA), Random Forest (RF), Naive Bayes (NB), Multi Layer Perceptron (MLP), Support Vector Machine (SVM), and a variation of the Artificial Neural Network class of methods. With a 99.5% accuracy rate, the Random Forest algorithm produced the best results.

The importance of developing cyber security has been recognized by academic and business institutions all around the world. Providing sustainable computing for the Internet of Things (IoT) is also crucial. IoT cybersecurity relies heavily on machine learning algorithms for malicious identification and intrusion detection. Therefore, in this work [2], the author (Fatani et al., 2022) leverages the benefits of swarm intelligence (SI) algorithms to create novel techniques for feature extraction and selection for the IDS system. The authors of this paper [2] (Fatani et al, 2022) approach to feature extraction relies on the use of traditional neural networks (CNN). Next, we introduce a different feature selection (FS) strategy that makes use of the recently created Aquila optimizer (AQU) SI algorithm. In addition, the quality of the created IDS technique was evaluated using four widely recognized public datasets: CIC2017, NSL-KDD, BoT-IoT, and KDD99. In order to confirm that the created method performs competitively, the authors (Fatani et al, 2022) also took into account thorough comparisons with alternative optimization techniques [2]. The outcomes demonstrate the developed approach’s excellent performance when measured using several assessment metrics.

The increasing number of IoT devices means that network security for IoT devices is critical. IoT devices with a variety of purposes make up IoT-based networks. Because of this variety of IoT devices, different dataset analysis

is required to guarantee that Machine Learning (ML)-based cybersecurity is implemented in an efficient manner. Still lacking, though, is the much-needed real-world IoT data needed for ongoing ML-based IoT data analysis. An extensive examination of the real-time IoT-23 dataset is presented in this research [3] by the author (Kim et al, 2022). A thorough examination of every one of the IoT-23 dataset's twenty scenarios shows that feature selection techniques and detection algorithms are consistently applied. The suggested machine learning-based intrusion detection system (ML-IDS) significantly increases detection accuracy— in certain cases, to 100%. The authors (Kim et al, 2022) work further shows that even with resource-constrained IoT processors, ML-IDS implementation is possible because the minimum number of features needed for a high detection rate of better than 99% is still 2 or 3.

The use of IoT devices is expanding quickly, and security is becoming more and more critical in our daily lives. In the past few years, numerous IoT intrusion detection datasets, including TON_IoT, BoT-IoT, and Aposemat IoT-23, have been made available as part of the effort to combat these security threats. Numerous machine learning-based IoT intrusion detection models were constructed using these datasets. In this work [4], the authors (Alani & Miri, 2022) have provided an explainable and efficient approach to identify the most useful universal characteristics from IoT intrusion detection datasets, which can be used to develop machine learning-based intrusion detection systems that are both highly accurate and efficient. After using the suggested approach on the TON_IoT, Aposemat IoT-23, and IoT-ID datasets, six universal network-flow features were chosen. After testing, the suggested approach yielded a high accuracy of 99.62% and a 70% prediction time reduction. In order to enhance comprehension of the classifier's functioning, a Shapley additive explanation was employed to elucidate the chosen features and demonstrate the explanation's congruence with contemporary attack methodologies.

The brain-computer interface is a system that facilitates communication between neuroprosthetic devices and the human brain. In order to efficiently gather hidden patterns in matching electroencephalogram (EEG) spectra and to rectify uncontrolled EEG fluctuations, the author (Tiwari & Chaturvedi, 2019) has done signal smoothing in their study [5]. The author (Tiwari & Chaturvedi, 2019) has achieved this by applying a Savitzky Golay function, which preserves spectral features with minimal distortion. The author (Tiwari & Chaturvedi, 2019) used the Principle Component Analysis (PCA) in conjunction with the Filter Bank Common Spatial Pattern (FBCSP) technique for feature selection. Using EEG signals, FBCSP generates 11352 features, which PCA progressively narrows down to the 185 most important features. The eXtreme Gradient Boosting (XGBoost) algorithm uses these features in the classification process together with appropriate node split criteria to maintain ideal tree tallness. The better performance of XGBoost in terms of reducing execution time (3.7 times faster in the training phase) and offering improved accuracy in comparison to previous results is concluded using a five-fold cross-validation approach. Compared to regularised common spatial patterns (78.01% accuracy), shift variance approximation (73.84% accuracy), and the Riemannian approach (74.77% accuracy), the authors (Tiwari & Chaturvedi, 2019) method improves classification accuracy (88.80%) by about 10%, 15%, and 25%, respectively. It also concludes that a better approximation is obtained by taking the noise level in EEG spectra into consideration beforehand.

The author (Dara & Tumma, 2018) of this study [6] examines the effects of feature extraction in deep learning methods like convolutional neural networks (CNNs). The goal of this work is to provide an emerging assessment of the real literature during the last five years on feature extraction techniques. Large-scale research and analysis in the field of feature extraction have become increasingly effective as application demand has grown. The author (Dara & Tumma, 2018) has provided a thorough analysis of deep learning in this paper [6], along with a description of certain current feature extraction techniques.

The author of this paper (Zebari et al, 2020) provides a thorough analysis of FS and FE within the context of DR in this study [7]. Additionally, each paper's specifics— such as the datasets, classifiers, algorithms and methodologies used— are thoroughly examined and summarised. In addition, a thorough analysis of every approach studied was conducted in order to identify trends among the authors (Zebari et al, 2020) of this paper [7], identify the method or ways that resulted in a large reduction of computational time, and choose the most accurate classifiers. Consequently, the various forms of both approaches have been examined and the results evaluated by the authors (Zebari et al, 2020) of this paper [7].

Intrusion Detection Systems (IDSs) and other detection and prevention technologies are necessary due to the numerous attacks on the Internet of Things (IoTs). In intrusion detection systems, anomaly detection is a difficult and time-consuming process. It necessitates the creation of a strong classifier model that is able to discern between various assault kinds. There will be significant variations in the detection performance of IDS when various input features are

given. Furthermore, the amount of network traffic and its multifaceted nature will make the classification procedure time-consuming. The author (Do et al, 2021) of this paper [8] focuses on data preparation and suggests an additional feature extraction technique for categorising attacks in Internet of Things networks. The characteristics that have the biggest effects on the classification model's accuracy for malicious IoT traffics are also highlighted by the author (Do et al, 2021) of this paper [8].

When it comes to early malware detection on devices with novel or unusual architectures, cross-architecture malware detection is crucial. For the same work on new and uncommon architectures, prior knowledge of malware detection on conventional ones can be transferred. The author (Tran Nghi Phu et al, 2019) of this paper [9] offers CFDVex, a feature selection technique for detecting cross-architecture malware, based on CFD and Vex intermediate representation. The suggested method's experimental evaluation on our sizeable IoT dataset produced positive outcomes for the detection of cross-architecture malware. With only Intel 80386 architecture data used to train the SVM model, the author's approach detected IoT malware for MIPS architecture samples with 95.72% accuracy and 2.81% false positive rate.

In order to minimise harm to IoT devices, the author (Jeon et al, 2020) of this study [10] suggests a dynamic analysis for IoT malware detection that can identify both known IoT malware and newly discovered, intelligently evolved variants of IoT malware. The DAIMD system analyses IoT malware dynamically in a nested cloud environment and trains it using a convolution neural network model. DAIMD extracts memory, network, virtual file system, process, and system call behaviours from IoT malware using dynamic analysis in a nested cloud environment. Convolution Neural Networks classify and train behaviour images of IoT malware by transforming the extracted and analysed behaviour data into images. By visualising and analysing the large volume of behaviour data produced by dynamic analysis, DAIMD helps reduce the quantity of malware that infects IoT devices.

The Internet of Things (IoT) envisions a world in which virtually everything is interconnected, providing on-demand access to information and services. IoT architecture typically consists of three main layers: perception layer, network layer, and application layer. To establish a robust and secure IoT ecosystem, different security measures are necessary at each of these layers. Security is paramount to ensuring the long-term viability of IoT technology. Addressing and resolving security issues is critical to realising the full potential of IoT. Researchers and experts have made significant efforts to address security challenges specific to IoT layers and devices by implementing appropriate security measures. In this paper [11], the authors (R. J. and V. S. M, 2022) provide a high-level perspective on the concepts, technologies, and security challenges in the IoT domain. The authors (R. J. and V. S. M, 2022) explore potential solutions to security problems and outline future directions for improving IoT security. Additionally, the survey [11] delves into a detailed analysis of the security challenges and threats inherent in IoT applications. The final part of the discussion [11] highlights various existing and emerging technologies designed to establish a high level of reliability in IoT applications.

3. Dataset Discussion and Preprocessing Methods

IoT-23 is a brand-new dataset of Internet of Things (IoT) device network traffic. In IoT devices, it has captured 20 malware executions and 3 benign IoT device traffic grabs. It includes 981934 entries and 31 features. Twenty network grabs are used to separate these cases. A distinct malware sample that runs on a Raspberry Pi and makes use of a range of protocols and activities represents each risky scenario in the dataset. The network traffic for the benign scenarios was collected using three distinct Internet of Things devices: a Philips HUE smart LED light, an Amazon Echo home intelligent personal assistant, and a Somfy smart doorlock. A necessary step before using any machine learning model is data preprocessing. Any flaws, noise, or inconsistency in the data must be addressed. Improper preprocessing of the data can have a major effect on the model's performance. The phases of data preprocessing are feature selection, transformation, and cleaning. The libraries Pandas, NumPy, Scikit-learn, and SciPy were utilized to conduct data preprocessing. In our dataset, there were a lot of null or missing values. Such values are not suitable for training our dataset. We have imputed the mean to the data in order to fill the null values. In many machine learning tasks, it is an essential preprocessing step that makes it possible for algorithms to operate efficiently with categorical variables in a numerical setting. One of the most important steps in getting data ready for machine learning models is by normalising it. By bringing all features to a same scale, the aim is to guarantee that each feature contributes equally to the model's training process. A popular method of normalisation is Min-Max Scaling. The values within

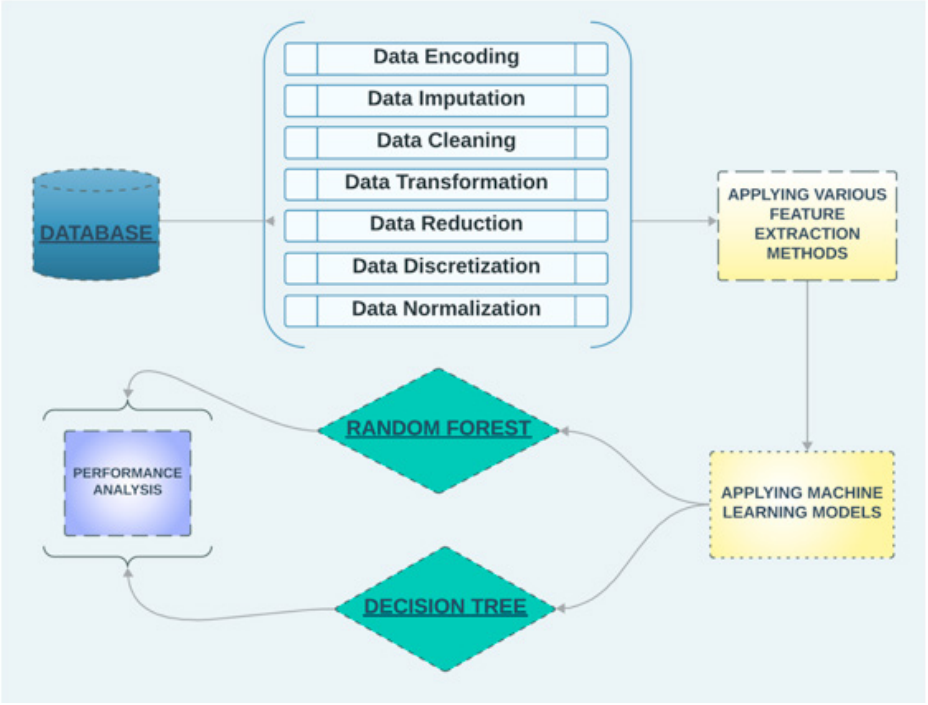


Fig. 1. (a) shows the Methodology .

each feature are changed by this method to fall between 0 and 1. As an alternative, we might think about standardising Z-scores using the StandardScaler class. By using this method, the characteristics are standardised to have a mean of 0 and a standard deviation of 1. The decision between Z-score standardisation and Min-Max Scaling frequently comes down to the specifications of the machine learning algorithm we want to utilise as well as the properties of our dataset. A more reliable and efficient learning process can be ensured by using the normalised dataset as input for machine learning models that are being trained and evaluated. Using Scikit-learn's StandardScaler, we were able to standardise the data by transforming it so that each feature's mean was equal to zero and its variance was one. It is a waste of computer time to train our model using duplicate values since we will be training the same data twice, which will not enhance any part of the model's performance.We use the Pandas package to avoid this, as it offers a straightforward `drop_duplicates()` method for eliminating duplicate rows from a DataFrame. Columns with mixed data types are common in real-world data-sets that we come across. These data-sets have numerical and category columns in them. Nevertheless, a number of machine learning models are incompatible with categorical data, thus this data must be transformed into numerical data in order to be fitted into the model. So for this we have done one hot encoding to our target feature. One-hot encoding is a machine learning technique that handles categorical variables by transforming them into a numerical representation that is simple for algorithms to understand. During this process, a binary vector— also known as a one-hot vector— represents each distinct category inside a categorical variable. The steps that are involved are not very complicated. First, every different category in the original variable is given a unique index or number name. Binary columns are then formed, each of which represents a distinct category. The binary column corresponding to each observation's category in the data set is set to 1, while all other binary columns are set to 0. As a result, each observation yields a sparse matrix with a single "hot" (1) member that denotes the existence of a particular category. By converting categorical variables into a binary format and generating new binary columns for each distinct category in the original feature, one-hot encoding functions. One-hot encoding preserves data regarding the existence or lack of particular categories. It able to manage several categories within of a feature by making distinct binary columns for every category.

4. Feature Extraction and Machine Learning methods

4.1. Mutual Information

Mutual Information calculates mutual information for continuous target variables in regression problems or fixed categories in classification tasks. The entropy of the variables is the foundation of mutual information. The mutual information between two random variables X and Y can be stated formally as follows:

$$I(X;Y) = H(X) - H(X|Y) \tag{1}$$

Where in equation (1) H(X) is the entropy for X, I(X; Y) is the mutual information for X and Y, H(X — Y) is the conditional entropy for X given Y. The outcome consists of bits in units of (zero to one).

4.2. XGBoost

XGBoost can compute feature importance scores, which indicate how much each feature contributes to the model's predictive performance. The higher the importance score, the more influential the feature is in making predictions. Feature importance scores are typically calculated based on how often a feature is used in decision trees and how much it contributes to reducing impurity in the data.

4.3. Correlation

The linear link between adjacent characteristics in a dataset is measured by the correlation matrix. It shows the degree and direction of a relationship between two features. The range of a correlation value is -1 to 1, with -1 denoting a strong negative connection, 0 denoting no correlation, and 1 denoting a high positive correlation.

4.4. Wrapper Method Random forest Using Gini Index

In order to extract features from our dataset using the Gini index as a wrapper method, a subset of those features will be chosen based on their Gini index values, which indicate how important those features are to a decision tree-based classifier.

$$Gini = 1 - \sum_{i=1}^{k} (p_i^2) \tag{2}$$

The impurity reduction attained by segmenting a dataset based on a specific attribute can be measured using the Gini index. For classification tasks, features that result in a bigger drop in Gini index when used as a split criterion are prioritised since they make it easier to tell one class from another.

4.5. Variance Threshold

Variance Threshold is a feature selector that removes all the low variance features from the dataset that are of no great use in modelling. It looks only at the features (x), not the desired outputs (y). The Default Value of Threshold is 0.

- If Variance Threshold =0 (Remove Constant Features).
- If Variance Threshold >0 (Remove Quasi-Constant Features).

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{(n-1)} \tag{3}$$

4.6. Decision Tree Machine Learning Model

A Decision Tree is a Supervised learning technique. A Decision Tree is set of axial hyper planes that divide the region into a hypercube. These can be used for both classification and Regression problems. In a Decision Tree, there are two nodes- decision node and leaf node respectively. A decision node is a work node where a flow branches into several optional flows. A leaf node is an abstract class for all nodes in a scene graph that have no children. Leaf nodes define lights, geometry and sounds; to provide special linking and display functions for sharing scenes; and provides a viewing platform for positioning and orienting the view in the virtual world. Decision Trees are used to get all possible solutions to a given problem. It classifies all records under specific class labels. The performance metrics used for decision tree are Gini Index, Entropy, and Classification Error. A measure of disorder or impurity in the provided data-set is called entropy. The decision tree divides the jumbled data according to the values of the feature vector that is connected to every data point. The data gets more homogeneous with each split, lowering the entropy. Nevertheless, not all data in all nodes will be homogeneous, meaning that the entropy number won't always be low. It is more difficult to make any conclusions the higher the entropy. Maximum purity is added when the tree eventually reaches the leaf node or terminal.

$$E = -\sum(p \cdot \log_2(p)) \tag{4}$$

Here in the equation (4) is p the probability and E is the entropy .

4.7. Random Forest Machine Learning Model

A Random Forest is a Supervised learning technique and can be used for both classification and Regression problems. Random Forest is based on the concept of ensemble learning. Here, instead of one decision tree, there are multiple decision trees from which the random forest takes prediction and based on the majority votes of predictions, it gives the final output. The basic math behind random forest is Based on a different random subset of the training set, Random Forest constructs several decision trees. By using bootstrapped sampling, each tree is trained on a random sample with replacements from the original dataset, resulting in the subsets. A random subset of features is taken into consideration for splitting at each decision tree node. By de-correlationing the individual trees, this randomness lessens over-fitting and enhances generalisation. There is some data that is not used in the training process for each tree because each tree is trained on a subset of the total data. The term "out-of-bag" (OOB) data refers to this. Without a separate validation set, the model's performance can be estimated using the OOB data. The average of each tree's predictions for regression or the majority vote for categorisation constitutes the Random Forest's final forecast. One point to note down is, the greater the number of trees in the random forest, the greater will be the chances of high accuracy. Random Forests evaluate the group's performance using a variety of metrics. Accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (AUC-ROC) are common measures used in classification. While precision and recall offer insights into the trade-off between correctly detecting positive examples and avoiding false positives, accuracy assesses the total correctness of predictions. The F1-score provides a balance between recall and precision. The model's capacity to discriminate across classes is assessed using AUC-ROC. Together, these indicators aid in assessing the Random Forest model's resilience and efficacy in a range of tasks. In Random Forest, regression metrics can also be used. For example, MAE, MAPE, MSE, RMSE, MASE and others. But mostly MAE and MSE are used.

$$\hat{y} = \frac{1}{N}\sum_{i=1}^{N} f_i(x) \tag{5}$$

Where in equation (5) y is the predicted output, N is the number of trees in the forest, f(x) is the prediction of the i-th tree for the input (x).

5. Examining Various Evaluation Metrics

5.1. Accuracy

A metric called accuracy assesses how accurate the model's predictions are overall. Its definition is the ratio of the total number of examples in the dataset to the number of accurately anticipated instances. Accuracy is calculated as the ratio of the number of correctly predicted instances to the total number of instances in the dataset. Here in the equation (6) a is accuracy, tp is Number of Correct Predictions and cp is Total Number of Instances.

$$A = \frac{tp}{cp}\% \tag{6}$$

Metrics like testing and training accuracy are used to assess how well a machine learning model is performing, frequently in the context of supervised learning. The model's capacity to accurately predict the labels of the training set of data is measured by training accuracy. It is calculated by contrasting the actual labels in the training data set with the predictions made by the model. Testing accuracy, on the other hand, evaluates how well the model generalises to new, untested data that it was not exposed to during training. The testing data set acts as a stand-in for actual situations, and testing precision aids in projecting the model's performance on novel, as-yet-undiscovered cases. The main distinction is in the data used for evaluation: testing accuracy measures the model's performance on independent data to provide a more realistic assessment of its ability to generalise, whereas training accuracy evaluates the model on the same data it learned from, potentially resulting in unduly optimistic estimates. Building reliable and efficient machine learning models requires striking a balance between high testing and high training accuracy. A model may be over fitting if it shows abnormally high accuracy, in this case the model has effectively memorised the training data rather than understanding its underlying patterns. Because the model has too closely fitted the noise and details of the training set, it may not generalise well to new, unknown data. Practical implementations may suffer from the model's inability to adjust to changes in real-world circumstances. Conversely, a model may be under-fitting if its accuracy is extremely poor, meaning that it is too simple to adequately represent the intricacy of the data. Due to a lack of complexity or an inadequate representation of underlying relationships, under-fitting models are unable to identify patterns in both the training and testing data sets. Finding the ideal balance between avoiding memorisation and oversimplification and creating a model that generalises well to new data is essential. To fine-tune models and reach an appropriate degree of accuracy that guarantees reliable performance across a variety of data sets, methods including as regularisation, cross-validation, and hyper-parameter tuning are utilised which we have discussed in our future area of work.

5.2. False Alarm

A false alarm, sometimes referred to as a Type I error or false positive, happens when a classification model predicts the positive class when the true class is negative. Stated differently, a false alarm occurs when the model incorrectly labels some events as positive when they ought to have been labelled as negative.False alarms are bad since they can generate needless actions or interventions when none is really warranted. The assessment and optimisation of classification models, such as decision trees, require careful consideration of the trade-off between reducing false alarms and guaranteeing the accurate identification of actual positive cases. A classification model's false alarm rate can be better understood and controlled by striking a balance between precision and recall measures.We must add up all of the false positives (FP) for every class and divide that amount by the total number of genuine negatives (TN) for every class in order to get the false alarm rate (FAR) for the Train Result. The formula to compute the false alarm rate is in the equation (7)

$$FAR = \frac{FalsePositives + TrueNegatives}{FalsePositives} \tag{7}$$

Precision evaluates how well the model predicts the favourable outcomes. The ratio of true positives to the total of true positives and false positives is used to compute it. Precision measures how well the model prevents declassifying

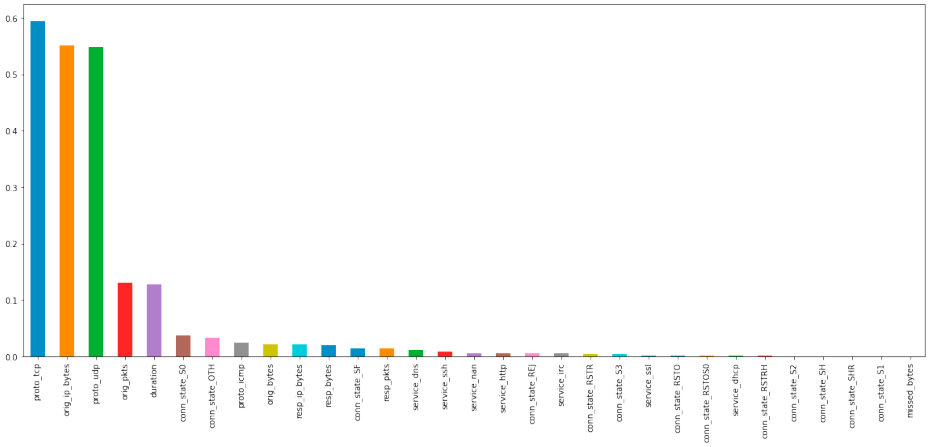


Fig. 2. (a)shows the plot between features and mutual Information score .

negative events as positive ones. A high precision indicates a high probability of accuracy when the model predicts a positive case. The formula to compute the Precision is in the equation (8)

$$Precision = \frac{TruePositives}{TruePositives + FalsePositives} \tag{8}$$

Recall, which is sometimes referred to as sensitivity or true positive rate, assesses a model's capacity to accurately detect every pertinent instance of a given class. The ratio of true positives to the total of false negatives and true positives is used to compute it. A high recall means that fewer good examples are likely to be missed because the model is able to accurately capture a sizable fraction of them. (FN) denotes false negative values in the data-set.

$$Recall = \frac{TruePositives}{TruePositives + FalseNegatives} \tag{9}$$

6. Results and Discussion

6.1. Findings

By selecting the most relevant features from the output of various feature extraction models, the assessment of the model was conducted. Subsequently, random forest and decision tree machine learning algorithms were utilized to train and test the selected features. and the models accuracy and false alarm rates have been computed to evaluate the models performance . Throughout the implementation process, a consistent procedure involved splitting the data, reserving 33% for testing purposes and allocating the remaining 67% for training. During the application of Mutual Information on our dataset, notable disparities were observed in the mutual gain for specific features. Utilizing XG-Boost for classifier-based feature extraction, we initially computed the feature importance values. Subsequently, we identified and plotted the top ten features. Upon analyzing the XGBoost feature importance, it became apparent that certain features held zero importance scores. Therefore, eliminating such features was deemed feasible, given their negligible impact on the dataset's performance. After computing all feature scores, we selected the best ten features and visually represented them in Figure 2.

When conducting variance threshold for feature extraction, the procedure involves computing variances for each feature and subsequently identifying and removing those with zero variance. This prioritizes the selection of the most crucial features. Utilizing SelectKBest, the goal is to retain the most informative features based on statistical scores. The resulting data frame contains features along with their variances. Furthermore, we have applied the SelectKBest method, which selects the best k features based on statistical scores. From the insights gained through our research on feature extraction techniques, findings from the xgb boost method were utilized. Decision tree and random forest

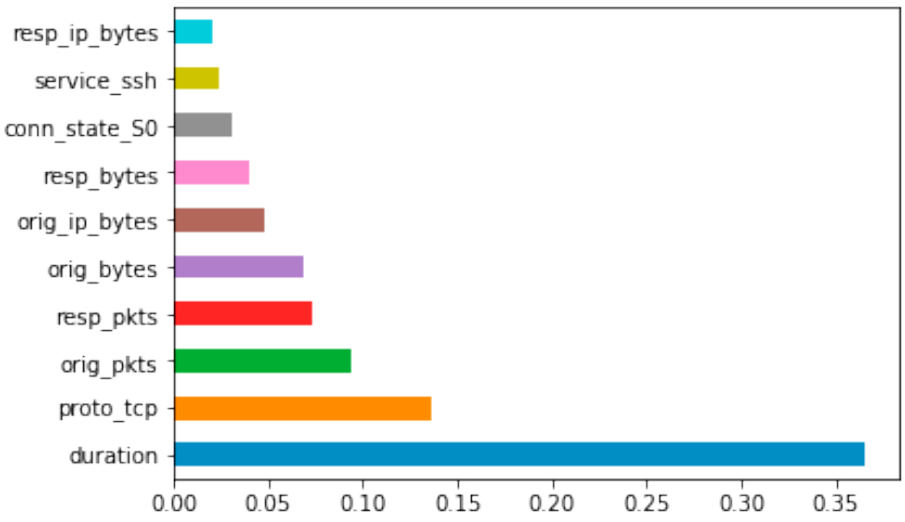


Fig. 3. (a)shows the plot between features and the feature importance score. .

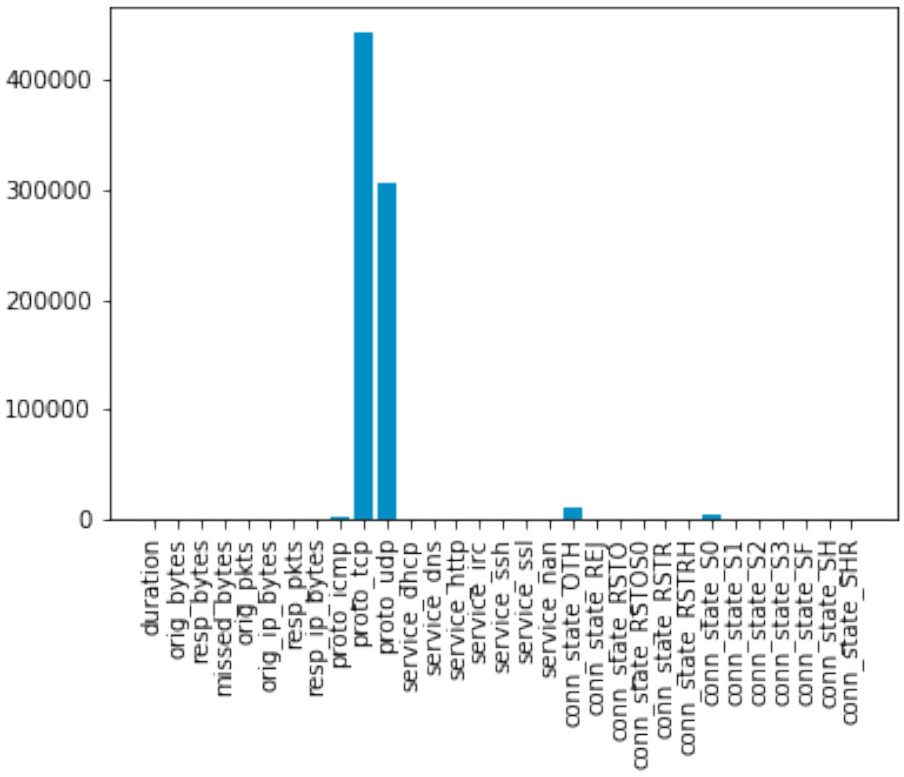


Fig. 4. (a)shows the plot between features and the k-best features score.

machine learning models were then implemented to evaluate the model's performance. By reducing the number of features to 26 using feature importance scores, the accuracy achieved after implementing decision tree was 92.98%, and for random forest with 10 decision trees, it was 93.02%. The false alarm rate for the training result using decision

tree for xgb boost-based feature extraction was 8.91%, while for the test result, it was 9.15%. Similarly, employing mutual information gain for feature extraction led to the selection of the top 26 features based on feature information gain. The resulting accuracy after implementing decision tree was 92.94%, and for random forest with 10 decision trees, it was 92.97%. The false alarm rate for the training result using decision tree for mutual information gain-based feature extraction was 6.20%, whereas for the test result, it was 6.26%. For the variance threshold feature extraction method, the top 24 features were chosen after calculating SelectKBest scores. The accuracy achieved after implementing decision tree was 93%, and for random forest with 10 decision trees, it was 93.01%. The false alarm rate for the training result using decision tree for variance threshold-based feature extraction was 9.44%, and for the test result, it was 9.45%. Additionally, employing the Wrapper method of Random Forest using the Gini index for feature extraction resulted in selecting the top 24 features based on Gini scores. The accuracy after implementing decision tree was 93.04%, and for random forest with 10 decision trees, it was 93.05%. The false alarm rate for the training result using decision tree for Wrapper method - Random Forest using Gini index-based feature extraction was 8.92%, and for the test result, it was 9.18%.

Conclusion and Future Area of Work

In short, the Internet of Things (IoT) is rapidly changing the way individuals interact and automating various aspects of daily life. With the increasing utilization of IoT devices, the demand for robust internet security to safeguard these devices from malicious activity is rising. This paper concentrates on the critical task of feature extraction from IoT data, particularly focusing on the widely used IoT 2023 dataset. The investigation explores and compares diverse feature extraction methods, spanning from traditional statistical techniques to machine learning-based approaches. The primary objective is to comprehend the characteristics of the IoT 2023 dataset and enhance its potential applications. Feature extraction assumes a pivotal role in efficiently managing vast, multidimensional IoT datasets. It diminishes dimensionality, rendering data more manageable and computationally cost-effective for subsequent IoT analytics tasks. The research conducted extensive experiments and provided an all-encompassing performance analysis of various feature extraction methods and their performance across different machine learning models. This analysis serves as a guide for selecting the most suitable method for specific IoT applications. Noteworthy findings reveal that reducing the number of features through feature extraction, particularly utilizing an XGBoost-based algorithm, insignificantly impacts the dataset's performance. Within the realm of the Internet of Things, where data quality and prediction accuracy hold paramount importance, feature extraction methods contribute to more efficient and informative IoT solutions. Ongoing efforts to harness the potential of IoT data continue through comprehending how feature extraction methods influence data quality, facilitating the development of intricate and secure IoT applications in 2023 and beyond. The horizon for further development in this work is an extensive subject. To narrow down its scope, we propose exploring the utilization of deep learning models in performance evaluation to enhance accuracy. Furthermore, efforts can be directed towards minimizing false alarm rates. Introducing hyper-parameter tuning can lead to improved model performance.

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